Distributed and Parallel Computing Project

Mini-Multiplayer Game

Joker Thief

Team Members

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The Game Concept

What is Joker Thief?

- **Concept**: A real-time multiplayer card game.
- **Players:** 2-8 humans or 1 human vs. 2-5 AI bots.
- **Deck:** Standard 52-card deck + 1 Joker (53 cards total).
- **Objective:** Avoid being the last player holding the Joker.
- **The Loser:** The player left with the Joker is "The Thief."

Player **Player**

OR

Player **Computer**

Core Mechanics & The Twist

How do you pair cards?

1 Classic games: Same Rank

2 Players automatically match to hand

Cannot be paired!

The Twist: Same Rank + Same Color Group

Red Pair

Receiver Flow

1 **Auto-Pair**

Players automatically match cards in hand

2 **Give Card**

Current player selects ONE card to pass

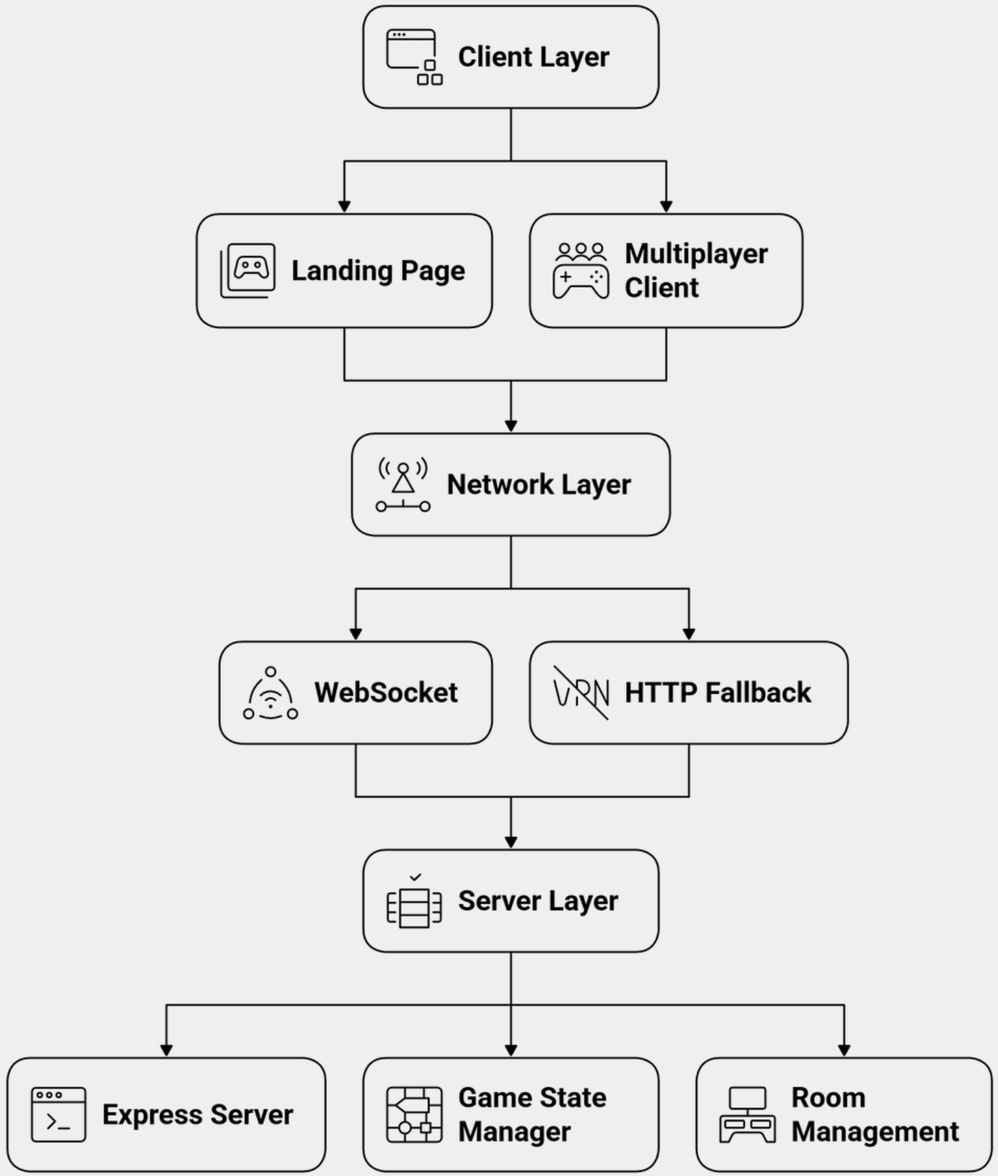
3 **Receive & Pair**

Next player receive and auto-pairs.

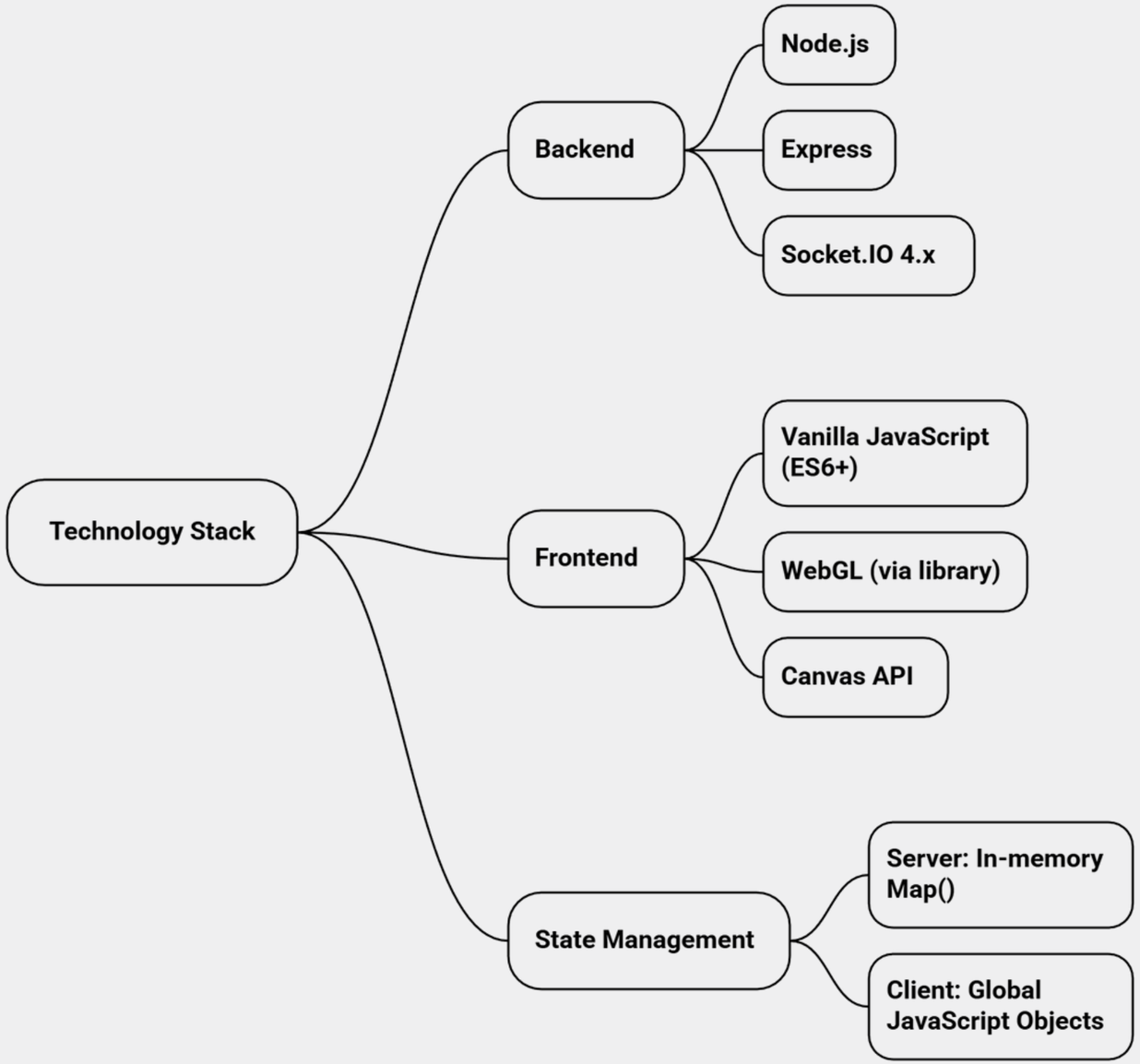
4 **Game End**

Only the Joker remains, Holder is "The Thief"

System Architecture



Technology Stack Using Websocket



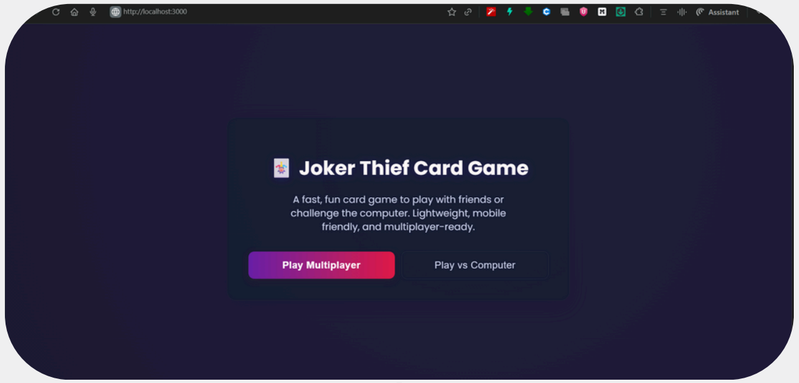
Event-Driven Architecture

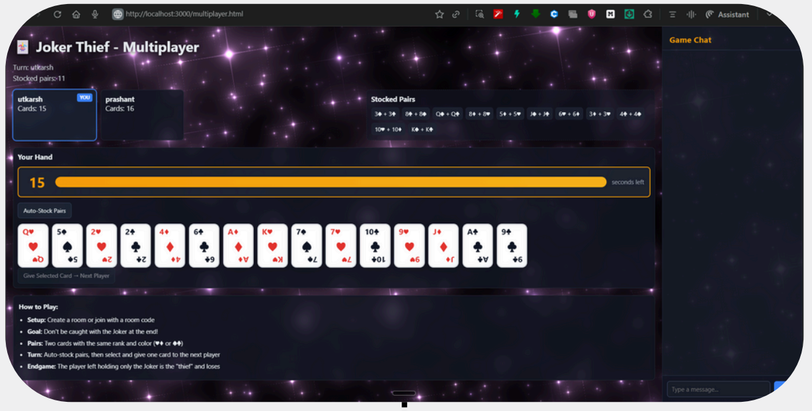
Event	Direction	Purpose
create-room	C → S	Initialize a new game room.
room-created	S → C	Confirm creation, send back roomId.
join-room	C → S	Join an existing room.
game-state	S → C	Broadcasts the public game state to all.
start-game	C → S	(Host only) Starts the game.
auto-pair	C → S	(Client-side logic, server validates)
give-card	C → S	The core game action. Pass card.
chat-message	Bi-directional	Real-time chat.

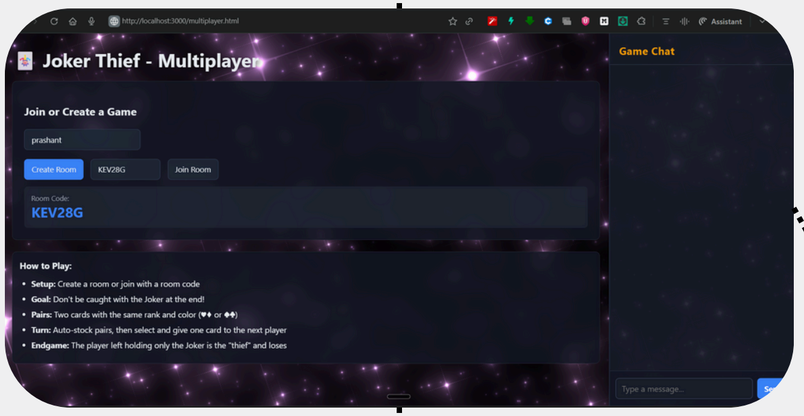
Core Socket Mechanisms

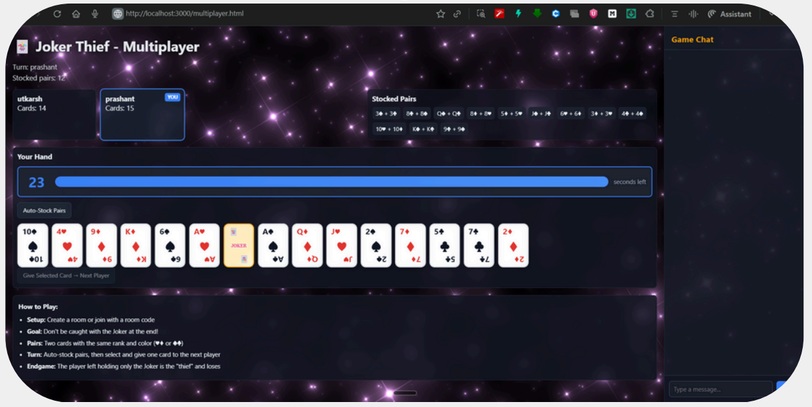
Component	How It Works	Example from Code
Event-Driven Messages	Named events carry JSON data	socket.emit('give-card', {index: 2})
Room Isolation	Each game is a separate "room"	socket.join('ABC123') + io.to('ABC123').emit(...)
Server Authority	All logic runs server-side; clients only request actions	Server validates turn, card index, and updates state
Selective Broadcasting	Public state to all, private hands to individuals	io.emit('game-state') + socket.emit('your-hand')

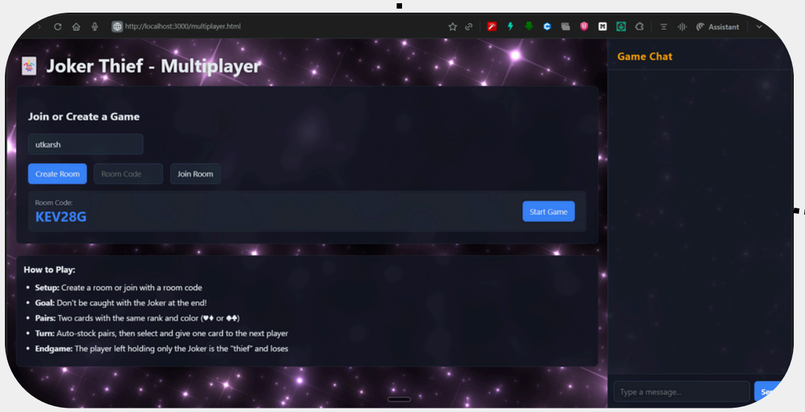
Chat Box

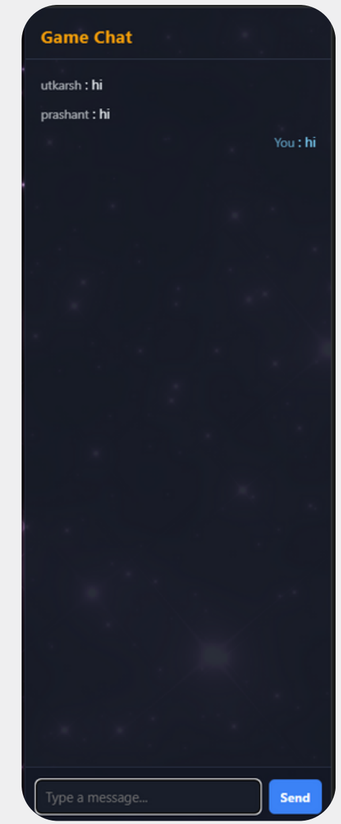













```
PS C:\Users\utkar\joker-thief-multiplayer> npm run dev
Player connected: SJnKBBq8kT2CFE3MAAAB
Player connected: JldNUBMIVpXe9GDJAAAD
```

Project Links & Demo

Click the below links

Project Resources

Live Game